

SOC 2026 : Final Project report

Skin disease classification model

1 Introduction

Skin diseases are among the most common medical conditions affecting millions of people worldwide. Early diagnosis plays a significant role in preventing severe complications, especially in diseases such as melanoma. This project develops a deep learning based image classification model using the ISIC 2019 Skin Lesion Dataset. The dataset was downloaded from the Kaggle image dataset website .

The model is implemented using the PyTorch framework and employs Transfer Learning through the ResNet18 architecture. The network learns discriminative image features and classifies skin lesion images into eight disease categories.

The trained model achieved an overall classification accuracy of **79.02%** with an average prediction confidence of **93.28%**.

2 Problem Statement

The objective of this project is to build an intelligent image classification AI Model capable of automatically identifying skin diseases from sample (or input)dermoscopic images. The model can classify images into one of the following skin disease variants:

1. Melanoma (MEL)
2. Melanocytic Nevus (NV)
3. Basal Cell Carcinoma (BCC)
4. Actinic Keratosis (AK)
5. Benign Keratosis (BKL)
6. Dermatofibroma (DF)
7. Vascular Lesion (VASC)
8. Squamous Cell Carcinoma (SCC)

3 Dataset Description

The project uses the ISIC 2019 Skin Lesion Dataset. The dataset consists of thousands of dermoscopic skin lesion images collected from multiple medical institutions. Each image belongs to one of eight disease classes.

The images are preprocessed before training by resizing them into a fixed resolution and converting them into tensors.

The dataset is divided into:

- Training Set
- Validation Set
- Testing Set

A total of 25331 images were used for training.

4 Utilities: PyTorch

PyTorch is an open-source deep learning framework developed by Meta AI. It provides Automatic differentiation (Autograd), GPU acceleration using CUDA, Neural Network module (torch.nn), Optimizers (torch.optim), Data loading utilities, Pretrained computer vision models. PyTorch represents images as multidimensional tensors. Every pixel value is converted into floating point numbers and stored inside tensors.

5 Training

Before training,

- Images are resized to fit a certain standard.
- Pixel values are normalized.
- Images are converted into tensors.

Initially, the neural network knows nothing about diseases. The convolution layers automatically learn properties like Edges Color variations, Texture, Shapes, Lesion boundaries and High-level disease features.

During every epoch, prediction errors are calculated using the loss function. Backpropagation computes gradients for every parameter. The optimizer updates weights to reduce prediction errors. This process repeats until the model converges.

No. of epochs used = 10

Epochs value can be increased for better accuracy while increasing the training time.

6 Model Architecture

The project uses Transfer Learning with ResNet18. The architecture consists of:

- Pretrained convolution layers

- Residual Blocks
- Global Average Pooling
- Fully Connected Layer

The final fully connected layer is replaced with:

Linear(512, 8)

to classify the eight skin diseases.

7 Training Pipeline

The training procedure follows these steps.

1. Load the dataset using DataLoader.
2. Load pretrained ResNet18.
3. Replace the final classification layer.
4. Move the model to GPU if CUDA is available.
5. Define CrossEntropyLoss.
6. Initialize Adam optimizer with

learning rate = 0.0001

- Forward Propagation
- Loss Computation
- Backpropagation
- Weight Update

8 Implementation

The code begins by importing essential PyTorch libraries. The pretrained ResNet18 model is loaded using torchvision.models. The final fully connected layer is modified according to the number of disease classes. Since training on images may also requires operations like convolutions, there is always some loss of data which is also present in the code as:

- Validation Loss
- Top-1 Accuracy
- Top-5 Accuracy

Finally, the trained model is evaluated on the testing dataset.

9 Experimental Results and evaluation metrics

The code also calculated and plotted graphs of evaluation metrics like Accuracy and precision. Recall, F1 Score and Average Confidence were also calculated for each class.

9.1 Overall Performance

Metric	Value
Overall Test Accuracy	79.02%
Average Prediction Confidence	93.28%
Epochs	10
Optimizer	Adam
Loss Function	CrossEntropyLoss
Architecture	ResNet18

9.2 Classification Report

Class	Precision	Recall	F1 Score	Support
MEL	0.71	0.63	0.67	862
NV	0.82	0.94	0.88	2616
BCC	0.82	0.76	0.79	720
AK	0.58	0.49	0.53	158
BKL	0.73	0.53	0.61	493
DF	0.79	0.44	0.57	52
VASC	0.82	0.84	0.83	49
SCC	0.59	0.43	0.50	116

Overall Accuracy = 79.02%

10 Future Scope

Future improvements include:

- Training deeper architectures such as ResNet50 or EfficientNet.
- Data augmentation.
- Hyperparameter optimization.
- Explainable AI using Grad-CAM.
- Deployment as a web application.
- Mobile application for real-time diagnosis.

11 Conclusion

This project successfully developed an automatic skin disease classification model using PyTorch and ResNet18. Transfer Learning significantly reduced training time while achieving an overall accuracy of 79.02%. The experimental results demonstrate that deep learning can effectively classify dermoscopic images and provide reliable predictions for medical image analysis.